

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Thermal Physics Lab **L-T-P:** 0-0-3 **Credits:** 2

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

1. Stefan's constant by using an incandescent lamp and a photo voltaic cell.
2. Thermal conductivity of a given disc of bad conductor of heat by Lees and Chorlton's method.
3. Thermo-e.m.f of a given thermocouple at different temperatures and the thermoelectric power.
4. Thermo-e.m.f of a given thermocouple with digital micro voltmeter at different temperatures and determination of the time constant.
5. Draw calibration curve of a platinum resistance thermometer and determine the boiling point of an unknown liquid using calibration curve.
6. Thermal expansion.
7. Heat Capacity of metals

Reference Books:

Lab manual